Predicting Customer Sentiment in Social Media Interactions: Analyzing AmazonHelp Twitter Conversations Using Machine Learning

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Abstract

Social media platforms generate large volumes of customer–service interactions, yet most sentiment analysis approaches focus on static classification of individual posts rather than capturing how sentiment evolves across conversations. This study addresses this limitation by proposing a machine learning framework to predict Conversation Polarity Change (CPC) in multi-turn Twitter interactions with AmazonHelp.

The proposed approach reconstructs conversation threads and labels sentiment transitions between the initial and final tweets as *Improved*, *Deteriorated*, or *No Change*. Textual features are extracted using TF-IDF, along with basic conversation-level characteristics, and multiple machine learning models are trained for CPC prediction.

Experimental results on 43,869 real-world Twitter conversations show that ensemble methods outperform traditional models. In particular, Random Forest achieves the best performance with an accuracy of 0.6247 and a macro F1-score of 0.5834, followed closely by Bagging. The results also reveal that positive sentiment transitions are easier to detect, while deteriorating sentiment remains more challenging due to its subtle and context-dependent nature.

These findings demonstrate that modeling sentiment at the conversation level provides deeper insights into customer experience compared to traditional post-level analysis. The proposed framework offers a practical approach for understanding sentiment dynamics in customer service interactions and can support the development of more responsive and intelligent support systems. The dataset consists of 787,346 conversations and over 2.8 million tweets.

1 Introduction

Social media platforms have transformed how organizations interact with customers, enabling real-time communication through platforms like Twitter. Corporate support accounts such as `@AmazonHelp` generate large volumes of conversational data that reflect customer concerns, responses, and overall brand perception (Pang and Lee, 2008; Medhat et al., 2014; Rosenthal et al., 2017; Ibrahim et al., 2017; Shirdastian et al., 2019; Arif et al., 2024).

Unlike traditional reviews, sentiment in social media is dynamic. A conversation may begin negatively and become positive after effective support, or worsen if unresolved. However, most sentiment analysis methods classify individual posts as positive, negative, or neutral (Pang and Lee, 2008; Go et al., 2009; Rosenthal et al., 2017), ignoring how sentiment evolves across multi-turn interactions.

Machine learning models such as Naïve Bayes, SVM, and Logistic Regression are widely used for sentiment classification (Go et al., 2009; Singla et al., 2017; Fitri et al., 2018; Yi and Liu, 2020). More recent approaches, including fastText and transformer-based models like BERT, improve contextual understanding (Joulin et al., 2016; Devlin et al., 2019; Kokab et al., 2022; Halawani et al., 2023; Mu et al., 2025). However, most studies still focus on single posts rather than conversation-level sentiment dynamics.

Social media analytics research also shows that customer interactions provide valuable insights into engagement and satisfaction (Ibrahim et al., 2017; Shirdastian et al., 2019; Park et al., 2021). Modeling how sentiment changes during interactions can therefore provide deeper insights than analyzing isolated posts.

To address this, this study proposes a machine learning framework to predict Conversation Polarity Change (CPC) in Twitter conversations with AmazonHelp. The goal is to determine whether customer sentiment remains stable or changes during an interaction. The main contributions are:

- Modeling sentiment at the conversation level using CPC.

- Comparing multiple machine learning algorithms.
- Designing tweet-, user-, and conversation-level features.
- Providing insights into how interaction patterns affect sentiment.

2 Related Work

Sentiment analysis has been widely studied in natural language processing, initially focusing on document- and sentence-level classification (Pang and Lee, 2008; Medhat et al., 2014). With the rise of social media, research shifted toward short and noisy text such as tweets (Go et al., 2009; Rosenthal et al., 2017).

Several studies have applied sentiment analysis in customer-related contexts. Lexicon-based and machine learning approaches have been used to analyze customer engagement and satisfaction, but they mainly rely on static classification (Saragih and Girsang, 2017; Al-Otaibi et al., 2020; Fitri et al., 2018). Similarly, sentiment-based recommendation systems improve decision-making but focus on individual posts rather than interactions (Singla et al., 2017; Yi and Liu, 2020).

Traditional models such as Naïve Bayes, SVM, and Logistic Regression remain effective, while fastText provides efficient text classification (Joulin et al., 2016). More recent work includes ensemble and real-time sentiment analysis systems (Xiao et al., 2023; Rahman et al., 2023). Deep learning models such as CNN and LSTM capture contextual information (Halawani et al., 2023; **?**), and transformer-based models like BERT further improve performance (Devlin et al., 2019). Hybrid and advanced models have also been explored (Kokab et al., 2022; Mu et al., 2025). However, these approaches still treat sentiment as a static problem.

Social media studies highlight the importance of interactions in understanding customer behavior and brand perception (Ibrahim et al., 2017; Shirdastian et al., 2019; Park et al., 2021). Although sentiment analysis is widely used in this context, only limited research focuses on conversation-level sentiment dynamics. A recent study using AmazonHelp data compared multiple machine learning models but simplified the problem by removing neutral and unchanged cases (Arif et al., 2024).

Therefore, modeling sentiment evolution across multi-turn interactions remains underexplored. This study addresses this gap by predicting Conversation Polarity Change (CPC), providing a more realistic representation of sentiment dynamics in customer service conversations. A comparison of existing sentiment analysis approaches is presented in Table 1.

3 Task and Dataset Description

This study focuses on modeling sentiment dynamics in customer service conversations on Twitter. The task is to predict *Conversation Polarity Change (CPC)* by analyzing multi-turn interactions between users and corporate support accounts such as `@AmazonHelp`. Specifically, the objective is to determine whether customer sentiment improves, deteriorates, or remains unchanged throughout a conversation.

The dataset is derived from the publicly available *Customer Support on Twitter* corpus, which contains large-scale interactions between customers and various brand support accounts. Each record includes tweet text, user information, timestamps, and conversation identifiers. Conversations are reconstructed using these identifiers to group all tweets belonging to a single interaction.

For each conversation, the first tweet represents the initial customer sentiment, while the final tweet reflects the outcome of the interaction. Sentiment labels are assigned to both using a sentiment classification model. Based on the transition between these two states, each conversation is labeled into one of three CPC categories:

- **Improved**: Negative → Positive
- **Deteriorated**: Positive → Negative
- **No Change**: Sentiment remains the same

To ensure meaningful analysis, conversations are filtered to include multi-turn interactions. The dataset contains conversations of varying lengths, reflecting real-world customer support behavior. The dataset statistics are summarized in Table 2.

The dataset exhibits typical social media characteristics, including informal language, abbreviations, and noise. Additionally, CPC class distribution is inherently imbalanced, as most interactions do not result in sentiment change. These challenges make the task more realistic and highlight the need for robust sentiment modeling techniques.

Overall, this dataset enables the study of sentiment evolution in real-world customer service interactions and supports the development of models

Study	Data	Method	Features	Limitations
Pang and Lee (2008)	General	Classical SA	Document-level features	Static sentiment only, no conversation modeling
Go et al. (2009)	Twitter	ML (NB, SVM)	Tweet-level features	Focused on single tweet classification
Singla et al. (2017)	Product Reviews	ML (NB, SVM, LR)	Text features	No multi-turn interaction analysis
Fitri et al. (2018)	Twitter	Naïve Bayes	Tweet sentiment features	Does not model sentiment transitions
Al-Otaibi et al. (2020)	Twitter	SVM	Unigram + bigram	Measures satisfaction, not sentiment change
Yi and Liu (2020)	Reviews	ML	Review sentiment features	Focused on recommendation systems
Xiao et al. (2023)	Social Media	ML + Meta-model	Text features	Real-time classification only
Kokab et al. (2022)	Social Media	BERT + CNN + BiLSTM	Contextual embeddings	Focused on post-level sentiment
Arif et al. (2024)	AmazonHelp Twitter	ML models	Conversation features	Removed neutral and no-change cases
Proposed Work	AmazonHelp Twitter	ML (LR, SVM, NB, KNN, DT, RF, Bagging)	TF-IDF + tweet, user, and conversation-level features	Models real-world Conversation Polarity Change (CPC) with improved, deteriorated, and no-change classes

Table 1: Comparison of existing sentiment analysis approaches and the proposed CPC-based framework

Metric	Value
Total Conversations	787,346
Total Tweets	2,811,774
Average Tweets per Conversation	3.57
Minimum Tweets per Conversation	2
Maximum Tweets per Conversation	Varies

Table 2: Dataset statistics

beyond traditional post-level sentiment classification.

4 Methodology

This study proposes a machine learning framework to predict Conversation Polarity Change (CPC) in Twitter-based customer service interactions. The pipeline consists of preprocessing, conversation construction and labeling, feature extraction, model training, and evaluation. The overall workflow is shown in Figure 1.

4.1 Preprocessing

Twitter data is inherently noisy and informal. To improve data quality, several preprocessing steps were applied. URLs, emojis, punctuation, special characters, and numbers were removed. All text was converted to lowercase, and common stopwords were eliminated.

These steps help normalize the text and reduce noise, making it more suitable for feature extraction and model training.

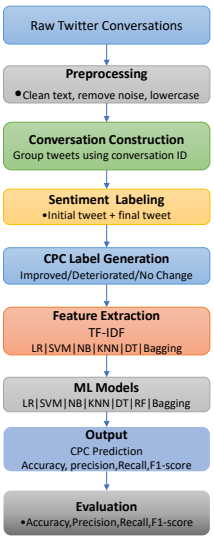

Figure 1: Proposed methodology for CPC prediction

4.2 Conversation Construction and Labeling

Conversations were reconstructed using conversation identifiers by grouping all related tweets into a single interaction thread. Each conversation consists of multiple exchanges between a customer and a support agent.

The first tweet represents the initial customer sentiment, while the final tweet reflects the outcome of the interaction. Sentiment labels were assigned to both using a sentiment classification

approach.

Based on the transition between the initial and final sentiment, each conversation was assigned a CPC label:

- **Improved**: Negative → Positive
- **Deteriorated**: Positive → Negative
- **No Change**: No polarity shift

This formulation enables modeling sentiment evolution rather than static sentiment classification.

4.3 Feature Extraction

Textual data was transformed into numerical form using TF-IDF vectorization with unigram features. TF-IDF captures the importance of words across conversations while reducing the impact of common terms.

In addition to textual features, basic conversation-level features such as the number of tweets per conversation were incorporated to reflect interaction dynamics.

4.4 Model Training

The dataset was split into training and testing sets to evaluate model performance. Several classical machine learning models were trained using the extracted features.

4.4.1 ML Models

The following models were used:

- Logistic Regression (LR)
- Support Vector Machine (SVM)
- Naïve Bayes (NB)
- K-Nearest Neighbor (KNN)
- Decision Tree (DT)
- Random Forest (RF)
- Bagging Classifier

These models were selected due to their effectiveness in high-dimensional text classification tasks and their ability to provide a strong comparative baseline.

4.4.2 Model Settings

TF-IDF features were used as input for all models. Regularization was applied in Logistic Regression and SVM to reduce overfitting. Random Forest was configured with multiple estimators to improve robustness. Most models were implemented with default parameters, with minor tuning where necessary.

4.5 Evaluation

Model performance was evaluated using accuracy, precision, recall, and F1-score. These metrics provide a balanced assessment, especially in the presence of class imbalance.

A comparative evaluation across models was conducted to identify the most effective approach for predicting sentiment changes in customer service conversations.

5 Experiments and Results

The performance of various machine learning models on the test set is presented in Table 3. From the results, it is clear that ensemble-based models outperform the other approaches for predicting Conversation Polarity Change (CPC).

Among all models, Random Forest achieved the best overall performance with the highest accuracy of 0.6247 and a macro F1-score of 0.5834. Bagging also showed very competitive results with a macro F1-score of 0.5823, which is very close to Random Forest. This indicates that ensemble methods are more effective in capturing sentiment dynamics compared to individual models.

Traditional models such as Logistic Regression and SVM produced moderate performance, with macro F1-scores of 0.5666 and 0.5553, respectively. Decision Tree showed slightly lower results, while KNN and Naïve Bayes performed comparatively worse. Overall, the results suggest that simple models struggle to capture complex sentiment transitions in multi-turn conversations.

Classifier	Accuracy	P	R	F
Logistic Regression	0.5699	0.5600	0.5889	0.5666
SVM	0.5711	0.5508	0.5677	0.5553
Naïve Bayes	0.5348	0.5220	0.4606	0.4232
KNN	0.5145	0.4880	0.4894	0.4853
Decision Tree	0.5532	0.5420	0.5492	0.5450
Random Forest	**0.6247**	**0.6078**	**0.5885**	**0.5834**
Bagging	0.6120	0.5938	0.5848	0.5823

Table 3: Performance of various machine learning models for CPC prediction, where P, R, and F denote precision, recall, and macro F1-score, respectively

The CPC dataset used in this study contains 43,869 conversations. Among them, *No Change* is the largest class with 42.15%, followed by *Improved* with 35.53% and *Deteriorated* with 22.33%. This imbalance has an impact on model performance, especially for the minority class.

A closer analysis of the best-performing Random Forest model shows that the *Improved* class achieved the strongest results, with a precision of 0.6816, recall of 0.8617, and an F1-score of 0.7611. In contrast, the *Deteriorated* class was more difficult to predict, with a recall of 0.3150 and an F1-score of 0.4025. The *No Change* class achieved a more balanced performance, with an F1-score of 0.5867.

These results indicate that positive sentiment transitions are easier to capture than negative sentiment deterioration. In customer support conversations, improvements are often expressed clearly, while deterioration tends to be more subtle and harder to detect.

Overall, the results show that CPC prediction is a challenging task. However, the strong performance of Random Forest and Bagging demonstrates that ensemble-based approaches are more effective for modeling sentiment evolution in social media customer service interactions.

6 Error Analysis

6.1 Quantitative Analysis:

We utilized a confusion matrix for error analysis of the best-performing model, Random Forest, as shown in Figure 2. The confusion matrix provides insight into how well the model distinguishes between the three CPC classes: 'Improved', 'No Change', and 'Deteriorated'.

From the confusion matrix, it can be observed that the 'Improved' class achieved the highest True Positive Rate (TPR). The model correctly classified 2686 instances of the Improved class, indicating that positive sentiment transitions are relatively easier to identify.

In contrast, the 'Deteriorated' class showed the lowest TPR. A significant number of deteriorated conversations (1139 instances) were misclassified as 'No Change'. This indicates that negative sentiment transitions are more subtle and difficult to detect compared to positive changes.

Furthermore, the 'No Change' class achieved a moderate level of performance, with 2178 correctly classified instances. However, a noticeable number

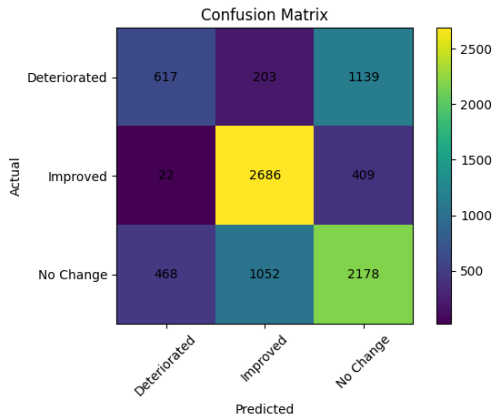

Figure 2: Confusion matrix of the Random Forest model for CPC prediction

of these instances were misclassified as 'Improved' and 'Deteriorated'. This suggests that the boundary between stable sentiment and slight sentiment variation is not clearly separable.

Overall, the confusion matrix highlights that the model performs well in identifying sentiment improvement but struggles to accurately capture deterioration and subtle sentiment stability.

6.2 Qualitative Analysis:

To further understand the misclassification behavior, we analyzed several example conversations. Figure 3 presents a few representative cases where the model predictions differ from the actual labels.

Text	Actual	Predicted
age arrive within days hit us saturday	Deteriorated	Deteriorated
ied customer service mins phone sort	No Change	Deteriorated
ermission yet tell security reasons wtf	Improved	Improved
r reorder amp waive day shipping fee	Deteriorated	No Change
ninutes alexa listen driving crazy turn	Improved	Improved

Figure 3: Sample predictions of the Random Forest model for CPC classification

From the qualitative analysis, several challenges can be identified. First, customer conversations often contain informal language, abbreviations, and noisy expressions. This makes it difficult for the model to accurately interpret sentiment, especially

when the expressions are ambiguous or context-dependent.

Second, many deteriorated conversations are expressed in subtle or indirect ways. For example, customers may express dissatisfaction without using explicitly negative words, which leads the model to misclassify such cases as 'No Change'. This explains the high confusion between the 'Deteriorated' and 'No Change' classes observed in the confusion matrix.

Third, the model primarily relies on the initial customer message to represent the conversation. However, sentiment evolution occurs across multiple interaction turns. As a result, important contextual information from intermediate messages is not fully captured, which can lead to incorrect predictions.

Finally, the dataset includes multilingual conversations, while the sentiment labeling method is based on an English lexicon. This mismatch can introduce errors in sentiment detection, particularly for non-English tweets, and consequently affect CPC classification performance.

Overall, these factors contribute to the observed misclassifications and highlight the complexity of modeling sentiment evolution in real-world customer service conversations.

7 Conclusion and Limitations

This study proposed a machine learning framework for predicting Conversation Polarity Change (CPC) in Twitter-based customer service interactions, using AmazonHelp conversations. By modeling sentiment evolution across multi-turn interactions, the approach provides a more realistic understanding of how customer perception changes compared to traditional post-level sentiment analysis.

Among the evaluated models, ensemble methods performed best. In particular, Random Forest achieved the highest macro F1-score of 0.5834, followed closely by Bagging. The model showed strong performance in identifying the *Improved* class, while the *Deteriorated* class remained more difficult to predict, indicating that negative sentiment transitions are harder to capture.

Despite these results, several limitations exist. The model relies mainly on the initial customer message, which does not fully represent the entire conversation. In addition, the dataset includes multilingual content, while sentiment labeling is based on an English lexicon, introducing potential noise. Informal and noisy social media text also affects prediction accuracy.

Future work can address these issues by incorporating full conversation context, using multilingual sentiment models, and applying advanced techniques for handling imbalanced data. Overall, CPC prediction remains a challenging task, and this work provides a useful baseline for future research on sentiment dynamics in social media interactions.

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